Extended Kalman Filter

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Where this fits:

- **Builds on:** Chi-square tests; Derivatives of one variable; Jacobian and matrix gradients; Kalman filter.
- **Compare with:** Particle filter.

1. Overview

Key point: Most robot sensors and motions are nonlinear functions of the state, and a nonlinear function bends a Gaussian out of shape. The extended Kalman filter replaces each nonlinear function by its tangent line at the current estimate and then runs the ordinary Kalman filter; checking its **innovations** (G-2511) tells us whether its variance can be trusted.

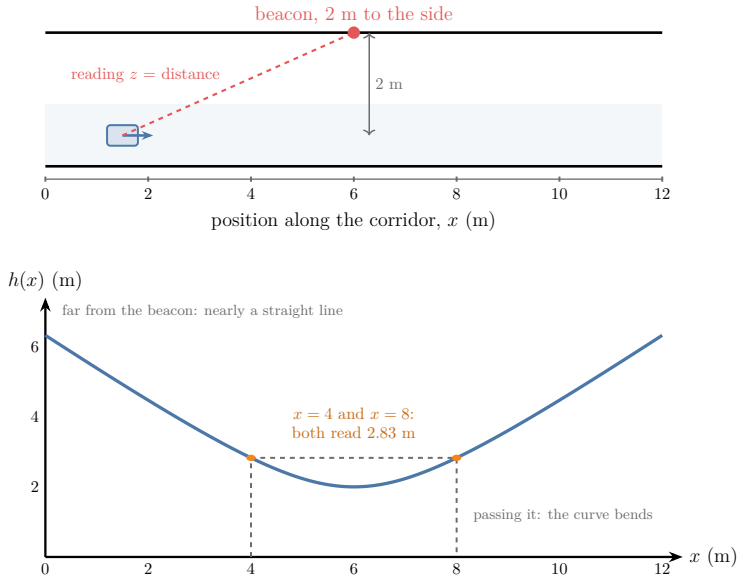


Figure 1: Top: a radio beacon on the corridor wall at 6 m, 2 m to the side; the robot reads its straight-line distance to it. Bottom: that distance as a function of position, a curve with its lowest point at 2 m under the beacon; positions 4 m and 8 m both read 2.83 m

With the **Kalman filter** (G-2506), the corridor robot blends its drifting commands and its jumping readings, and its estimate misses by 0.17 m, against 0.27 m for the commands alone and 0.51 m for the readings. The blend works because the belief stays a **Gaussian** (G-827), a bell-shaped curve set by a mean and a variance, and because the ceiling system reports the position itself plus noise, a **linear function** (G-2527) of the position, whose graph is a straight line. The robot now has a new sensor in the same corridor: a radio beacon fixed on the wall at 6 m along the corridor, 2 m to the side, and it measures its straight-line distance to the beacon (Figure 1). Such range sensors are common: radio beacons, GPS ranges to satellites, a lidar's range to a landmark.

The reading is no longer the position plus noise. Its graph against the position is a curve

(Figure 1), so no fixed multiple of the position plus a constant gives it: the reading is a **nonlinear function** (G-2530) of the position:

- far from the beacon, the distance changes almost one for one with position, nearly a straight line;
- under the beacon, the distance stops changing: it reaches its minimum of 2 m;
- past the beacon, it grows again, so 4 m and 8 m give the same reading.

The Kalman filter needs linear models (Section 2), so it cannot use this reading as it is. This Note adds the extended Kalman filter, which straightens the curve at the current estimate, and a check that detects when the result is wrong:

- pushing a Gaussian through a nonlinear function, and the tangent-line fix (Section 3);
- the extended Kalman filter, step by step on the corridor (Section 4);
- its limits: one Gaussian can hold only one side (Section 4.4);
- consistency, whether the filter's variance matches its real errors: the innovation, its covariance, and the NIS and NEES tests (Section 5).

2. The Kalman filter in brief, and why it fails for nonlinear functions

> **Key point:** The Kalman filter is exact only because **linear functions** (G-2527), whose graphs are straight lines, keep a Gaussian a Gaussian. A nonlinear function, such as the beacon distance, does not.

To see why the nonlinear beacon distance of Figure 1 cannot be used as it is, we recall what the **Kalman filter** (G-2506) does. The Kalman filter keeps the belief as a mean $\hat{x}$ and a variance p and runs two steps per round:

- **predict:** move the mean by the command, 0.5 m, and add the process noise Q, the variance of each step's wheel slip, 0.01 m^2, to the variance;
- **update:** blend in the reading z with the Kalman gain (G-2508), which gives the reading the weight that makes the new variance smallest.

Both steps rest on one fact: a linear function keeps a Gaussian a Gaussian, with the variance multiplied by the slope squared. The update also assumed the reading was a linear function of the position: a fixed number H times the position, plus the sensor's noise. Our new reading is the straight-line distance to the beacon, 6 m along the corridor and 2 m to the side. With:

- x is the robot's position along the corridor, in m;
- $h(x)$ is the distance from position x to the beacon, the reading before the sensor's noise, in m.

By Pythagoras, from the gap along the corridor and the 2 m across:

$$h(x) = \sqrt{(6 - x)^2 + 2^2}$$

For example, at 0.5 m:

$$h(0.5) = \sqrt{5.5^2 + 4}$$

$$= \sqrt{34.25} = 5.852 \text{ m}$$

No single number H turns position into this reading everywhere, so the five lines of the Kalman filter cannot be used as they are. The first guess is that of the Kalman filter: the estimate $\hat{x}$ is 0 m, with variance p of 1 m^2. The beacon's distance readings have a standard deviation of 0.1 m, so the **measurement noise** (G-2510) is $R = 0.01$ m^2. Before the filter can use these readings, we need to know what the beacon distance does to a **Gaussian** (G-827) belief, a bell-shaped curve over positions.

3. Pushing a Gaussian through a nonlinear function

> **Key point:** A **nonlinear function** (G-2530), one whose graph is a curve, bends a Gaussian into a lopsided shape. Near a point, its graph looks like its tangent line, and a line keeps a Gaussian a Gaussian; that approximation is good when the Gaussian is narrow compared with the bend.

3.1 What a nonlinear function does to a Gaussian

The worst place is near the beacon, where the distance reaches its minimum of 2 m (Figure 1). Suppose the robot's position there is a **Gaussian** (G-827), a bell-shaped curve, with mean 5.5 m and standard deviation (sd) 0.5 m. What does the predicted reading look like? We cannot use a formula yet, so we use the Monte Carlo way: draw 200,000 positions from the Gaussian, push each through the beacon distance h, and look at the results (notebook).

Figure 2 shows the result in its frame with input mean 5.5 m, sd 0.5 m. The readings concentrate just above 2 m, the smallest possible distance, with a long tail upwards: the Gaussian has become lopsided. Its lopsidedness, the **skewness** (G-1817), is 1.89, where a Gaussian has 0. A **Kalman filter** (G-2506) cannot store that shape: it keeps only a mean and a variance. To keep the belief Gaussian, the filter needs a straight line in place of the curve.

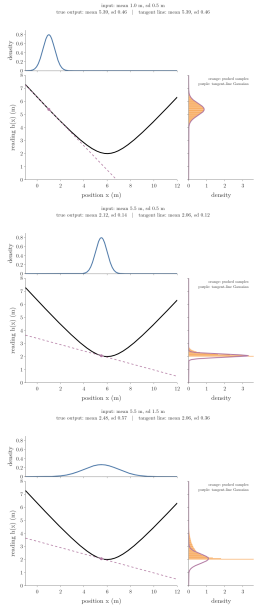


Figure 2: A Gaussian of positions (top) pushed through the beacon curve (middle, black) gives the orange histogram of readings (right); the purple Gaussian is what the tangent line at the mean (purple dashed) predicts. The input's mean steps from 1 m through 2.5 m and 4 m to 5.5 m, and then the input widens. Far from the beacon (mean 1 m) the two agree; near it (mean 5.5 m) the true readings are lopsided against 2 m, and with a wide input (sd 1.5 m) the tangent-line Gaussian is far too narrow

3.2 The tangent line: linearisation

Any smooth curve looks straight when viewed closely enough. Near a point, the curve is close to its tangent line, the straight line that touches the curve there with the curve's slope. Figure 3 zooms in on the beacon curve at two places at once, the window shrinking from 8 m to 0.1 m wide. Watch the gap between the curve and its tangent line: at 1 m it is already too small to see when the window is 2 m wide, while at 6 m, under the beacon, where the curve bends most, it is still visible at 0.5 m wide. So the tangent line can stand in for the curve close to where it touches, and the more the curve bends, the closer that has to be.

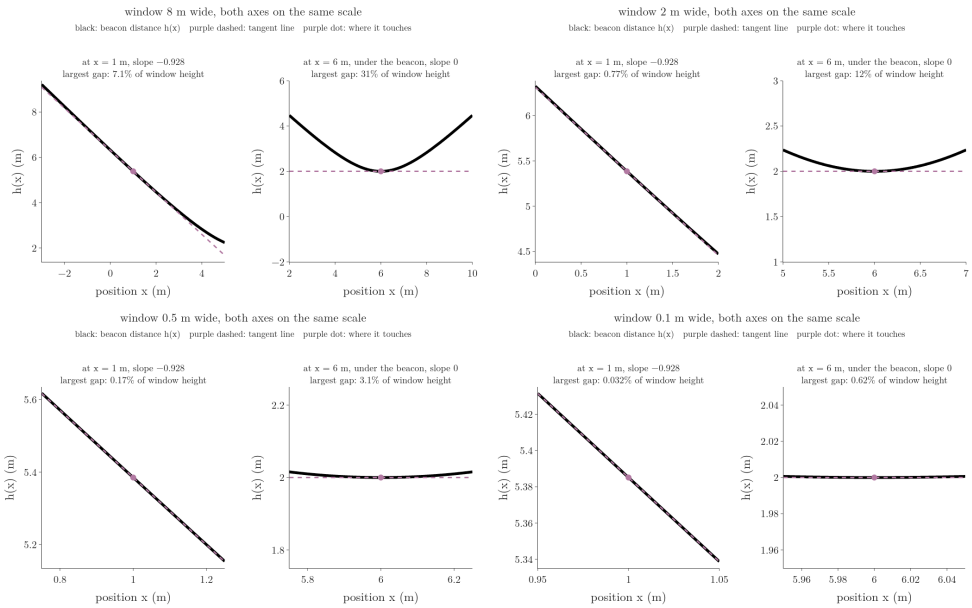


Figure 3: The beacon distance h(x) (black) and its tangent line (purple dashed) at x = 1 m (left, slope −0.928) and at x = 6 m under the beacon (right, slope 0), seen through a window that shrinks from 8 m through 2 m and 0.5 m to 0.1 m wide, with both axes on the same scale. Each panel gives the largest gap between curve and line inside the window, as a share of the window's height: at 1 m it falls from 7.1% to 0.77%, 0.17% and 0.032%; at 6 m from 31% to 12%, 3.1% and 0.62%. The last frame, 0.1 m wide: in both panels the curve cannot be told from its tangent line

Replacing a function near a point by its tangent line is linearisation (G-1099). In Figure 2 the tangent line is the purple dashed line, drawn at the input's mean. With:

- μ is the position where the line touches the curve, in m (1 m in the example below);
- $h(\mu)$ is the beacon distance there, in m;
- $h'(\mu)$ is the **derivative** (G-595) of h at μ, the slope of the curve there, in metres of distance per metre of position;
- x is a position near μ, in m.

$$h(x) \approx h(\mu) + h'(\mu)\,(x - \mu)$$

The beacon distance is the square root of an inside part, so the chain rule (G-371) gives its slope: the slope of the square root times the slope of the inside part. With:

- $s(x)$ is the part under the square root, in m^2, 29 m^2 at 1 m;
- dh/ds is the slope of h against s, in m per m^2, 0.0928 at 1 m;
- ds/dx is the slope of s against x, in m^2 per m, −10 at 1 m.

At 1 m first, each line followed by the same step at any position x:

$$s(1) = (6 - 1)^2 + 4 = 29$$

$$s(x) = (6 - x)^2 + 4$$

$$h(1) = \sqrt{29} = 5.3852$$

$$h(x) = \sqrt{s(x)}$$

$$\frac{dh}{ds} = \frac{1}{2 \times 5.3852} = 0.0928$$

$$\frac{dh}{ds} = \frac{1}{2\sqrt{s}} = \frac{1}{2h(x)}$$

$$\frac{ds}{dx} = -2 \times (6 - 1) = -10$$

$$\frac{ds}{dx} = -2\,(6 - x)$$

$$h'(1) = 0.0928 \times (-10) = -0.928$$

$$h'(x) = \frac{1}{2h(x)} \times (-2)\,(6 - x)$$

$$= \frac{-(6 - x)}{h(x)}$$

Driving 1 m towards the beacon from 1 m cuts the distance by about 0.93 m.

The tangent line is a straight line, and a straight line keeps a **Gaussian** (G-827) a Gaussian. So the predicted reading is a Gaussian whose mean is $h(\mu)$ and whose standard deviation (sd) is the input's sd times the size of the slope, where $|h'(\mu)|$ is the slope without its sign:

$$\text{mean} \approx h(\mu)$$

$$\text{sd} \approx |h'(\mu)| \times \text{sd of } x$$

Carrying a mean and a standard deviation through a function this way, with its value and slope at the mean, is **first-order uncertainty propagation** (G-2519) (Labbe ch.9; Becker ch.13).

3.3 When the tangent line fails

The tangent line describes the curve well only near the point where it touches. The wider the **Gaussian** (G-827), and the more the curve bends inside it, the worse the approximation (Labbe ch.9; Freiburg EKF slides). Three of the frames of Figure 2, with the notebook's numbers:

Input Gaussian	True readings: mean, sd	Tangent line: mean, sd
mean 1.0, sd 0.5 (curve nearly straight)	5.39, 0.46	5.39, 0.46
mean 5.5, sd 0.5 (near the bend)	2.12, 0.14	2.06, 0.12
mean 5.5, sd 1.5 (near the bend, wide)	2.48, 0.57	2.06, 0.36

Near the bend, the tangent-line Gaussian puts its mean too low and is too narrow, so a filter using it would be overconfident. The tangent-line Gaussian even gives probability to readings below 2 m, which no position can produce. Both failures grow with the width of the input: with standard deviation 1.5 m, the tangent line misses the true standard deviation by more than a third.

3.4 Several variables: the Jacobian

The beacon distance has one input, the position, so one slope is enough. A robot's **pose** (G-2280), where it is and which way it faces, has several numbers: its position x and y on the floor, in m, and its heading θ, in radians. A function of several numbers has a slope for each pair of output and input. The table of all these slopes is the Jacobian (G-980), and it plays the part of the slope: near a point, the function behaves like multiplying by that matrix.

For motion the Jacobian is called G, the **motion Jacobian** (G-2522), and it carries the covariance forward as the state transition matrix F does in the Kalman filter's matrix form. With:

- P is the estimate's covariance matrix, one row and one column per state variable: the variances on the diagonal and the covariances off it (for the corridor's one number, the variance p);
- G is the motion Jacobian, one row per new state variable and one column per old one (for the corridor, the single number 1);
- G^{T} is its transpose, G flipped over its diagonal.

The carried covariance is:

$$GPG^{\mathsf{T}}$$

With the tangent line for one number and the Jacobian for several, the **Kalman filter** (G-2506) can now run on nonlinear models.

4. The extended Kalman filter

> **Key point:** The EKF predicts and updates with the nonlinear functions themselves, and uses their slopes at the current estimate wherever the **Kalman filter** (G-2506) used its state transition matrix F and measurement matrix H.

4.1 The algorithm

The **extended Kalman filter** (EKF) (G-2520) uses that tangent line to run the **Kalman filter** (G-2506), which keeps a Gaussian belief as a mean and a covariance, on a nonlinear model such as the beacon distance: it linearises the nonlinear models at the latest estimate, every step (Freiburg EKF slides; Thrun §3.3; Labbe ch.11). The Kalman filter column of the table below uses the symbols listed in its matrix form; bold letters are vectors. The EKF adds:

- g is the motion model, the function that gives the predicted state from the command u and the last estimate $\hat{\mathbf{x}}$ (on the corridor, the position plus 0.5 m);
- G is the **Jacobian** (G-980) of g at the last estimate $\hat{\mathbf{x}}$, a matrix with one row per new state variable and one column per old one (on the corridor, the single number 1);
- h is the sensor model, the function that gives the reading expected from a state (on the corridor, the beacon distance of Section 2);
- H is the Jacobian of h at the prediction $\bar{\mathbf{x}}$, the **measurement Jacobian** (G-2521), with one row per reading and one column per state variable (on the corridor, one number, -0.940 at step 1).

Each is linearised where the filter's best guess is at the moment it is used. The lines:

Step	Kalman filter	Extended Kalman filter
predict mean	$\bar{\mathbf{x}} = F\hat{\mathbf{x}}$	$\bar{\mathbf{x}} = g(u, \hat{\mathbf{x}})$
predict covariance	$\bar{P} = FPF^{\mathsf{T}} + Q$	$\bar{P} = GPG^{\mathsf{T}} + Q$
innovation	$z - H\bar{\mathbf{x}}$	$z - h(\bar{\mathbf{x}})$
innovation covariance	$S = H\bar{P}H^{\mathsf{T}} + R$	same, H = slope of h at $\bar{\mathbf{x}}$
gain	$K = \bar{P}H^{\mathsf{T}}S^{-1}$	same
update	$\hat{\mathbf{x}} = \bar{\mathbf{x}} + K(z - H\bar{\mathbf{x}})$	$\hat{\mathbf{x}} = \bar{\mathbf{x}} + K(z - h(\bar{\mathbf{x}}))$
	$P = (I - KH)\bar{P}$	same

The means go through the true **nonlinear functions** (G-2530); only the covariances and the gain use the slopes. If the models are linear, $G = F$ and $h(\bar{\mathbf{x}}) = H\bar{\mathbf{x}}$, and the EKF is the Kalman filter (Freiburg EKF slides).

In our corridor the motion is still a straight line, so $G = 1$ and the predict step is unchanged. Only the update needs the slope.

4.2 Step 1 by hand

On the corridor every symbol is one number. At step 1, with the step number as a subscript:

- $\bar{x}_1$ is the predicted position, in m;
- $\bar{p}_1$ is its variance, in m^2;
- $h(\bar{x}_1)$ is the reading the prediction expects, the beacon distance at $\bar{x}_1$, in m;
- H is the measurement Jacobian, the slope of the beacon distance at $\bar{x}_1$, in metres of distance per metre of position;
- z_1 is the beacon's reading, 5.61 m;
- S is the innovation variance, the variance the filter expects of the innovation, in m^2;
- K is the Kalman gain, which turns the innovation into a correction of the position, in metres of position per metre of distance;
- $\hat{x}_1$ is the updated estimate, in m;
- p_1 is its variance, in m^2.

Predict, as in the **Kalman filter** (G-2506), from the first guess, 0 m with variance 1 m^2, with the command 0.5 m and the slip's variance Q, 0.01 m^2:

$$\bar{x}_1 = 0 + 0.5 = 0.5 \text{ m}$$

$$\bar{p}_1 = 1 + 0.01 = 1.01 \text{ m}^2$$

Linearise the sensor at the prediction. The expected reading and the slope, kept to four digits because the variance below subtracts two nearly equal numbers:

$$h(0.5) = 5.852$$

$$H = -5.5/5.852 = -0.9398$$

Innovation. The **innovation** (G-2511) is the gap between the reading and the reading the prediction expected. The reading is $z_1 = 5.61$ m:

$$z_1 - h(\bar{x}_1) = 5.61 - 5.852 = -0.242$$

The robot is 0.24 m closer to the beacon than predicted.

Gain. First the innovation variance, with the sensor's variance R, 0.01 m^2:

$$S = 0.9398^2 \times 1.01 + 0.01$$

$$= 0.892 + 0.01 = 0.902$$

then

$$K = 1.01 \times (-0.9398)/0.902$$

$$= -1.0523$$

The gain is negative because the slope is: a shorter distance than expected means the robot is further along, so a negative innovation must move the estimate forward. Its size above 1 is expected too: it converts metres of distance into metres of position, and here the distance changes a little less than one for one.

Update:

$$\hat{x}_1 = 0.5 + (-1.0523)(-0.242)$$

$$= 0.5 + 0.255 = 0.755 \text{ m}$$

$$p_1 = (1 - (-1.0523)(-0.9398)) \times 1.01$$

$$= 0.01105 \times 1.01 = 0.0112 \text{ m}^2$$

The true position is 0.73 m. **A check.** After a reading this precise, the position should be known about as well as the sensor allows, which is the sensor's standard deviation divided by the slope:

$$\text{sd} = 0.1/0.94 = 0.106 \text{ m}$$

$$\text{sd}^2 = 0.106^2 = 0.0112 \text{ m}^2$$

matching p_1. The slope 0.94 set that precision, and the slope changes as the robot drives towards the beacon.

4.3 Sixteen steps, past the beacon

Figure 4 runs 16 steps, as the robot drives from 0.73 m to 7.86 m, past the beacon at 6 m. The animation plays step 1 one move per frame, with the numbers of Section 4.2, then each later step in two frames. In the top panel, watch the purple tangent line touch the beacon curve at each prediction: its slope H is the red dot added to the bottom panel. As H nears 0 at the beacon, the band around the estimate in the middle panel widens.

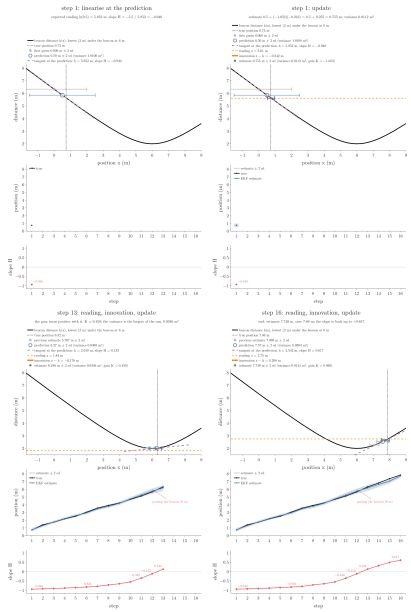


Figure 4: Top, one step at a time: the beacon distance h(x) (black), the prediction on it (hollow blue, ± 2 standard deviations), the tangent line there (purple, its slope H), the reading (orange line), the innovation (orange bar) and the updated estimate (blue dot). Middle: the EKF estimate follows the truth for 16 steps, with a band of two standard deviations that widens as the robot passes the beacon. Bottom: the slope H of the distance at each prediction rises from −0.94 through 0 at the beacon to +0.62. The last frame shows the whole run

The slope, the gain and the variance at six of the steps:

Step	Prediction (m)	Slope H	Gain K	Variance after (m^2)	Estimate (m)	True (m)
1	0.50	−0.940	−1.052	0.0112	0.755	0.73
6	3.08	−0.825	−0.662	0.0080	2.955	3.04
11	5.25	−0.349	−0.623	0.0178	5.270	5.13
12	5.77	−0.115	−0.307	0.0268	5.767	5.67
13	6.27	0.133	0.459	0.0346	6.186	6.32
16	7.57	0.617	0.868	0.0141	7.749	7.86

- Far from the beacon the slope is near −1: each reading locates the position precisely, and the variance stays between 0.007 and 0.011 m^2.
- Near the beacon, where the distance reaches its minimum of 2 m (Figure 1), the slope goes to 0: the distance hardly changes with position, so a reading says almost nothing about where the robot is. The variance grows to 0.035 m^2 at step 13, and the band in Figure 4 widens; the filter relies on its commands.
- Past the beacon the slope turns positive and the gain changes sign with it; the readings locate the position precisely again.

The filter needs no special case for any of this: recomputing H at every prediction is sufficient, as long as the filter starts on the right side of the beacon.

4.4 The limits of the EKF

One Gaussian holds one side. Positions 4 m and 8 m give the same reading, 2.83 m (Figure 1); so do 1 m and 11 m. Figure 5 feeds the same 16 readings, one step per frame, to a second EKF that starts at 11 m, on the far side of the beacon. Every reading fits the mirror position as well as the true one, so the filter follows the mirror path, confident and wrong: at step 1 it estimates 11.24 m when the robot is at 0.73 m, 10.5 m off, with only about 10 cm of uncertainty. Watch it move left while the robot moves right: only when the robot passes the beacon, where the mirror and the true path meet, does it become correct (7.96 m against the true 7.86 m at step 16, notebook).

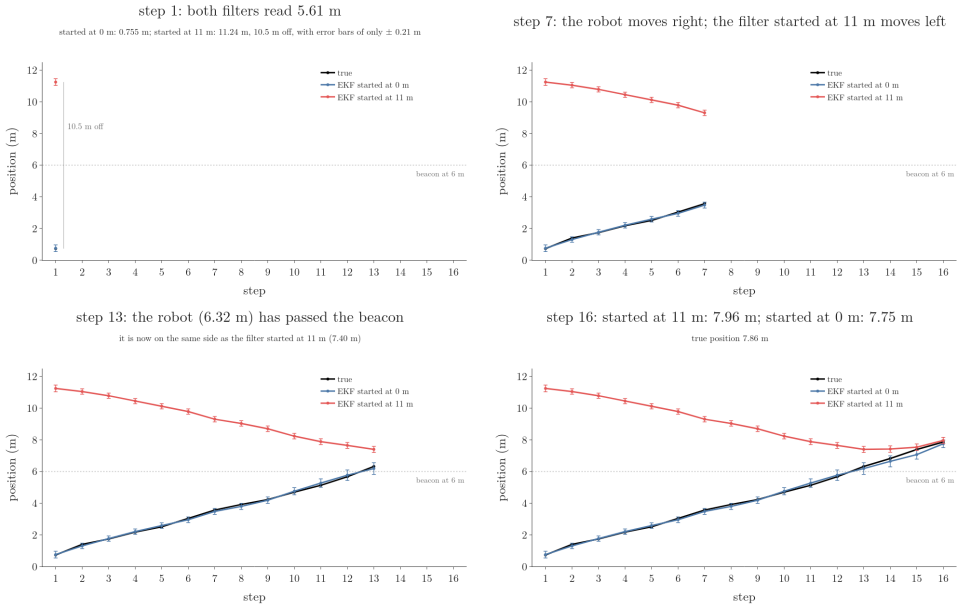


Figure 5: The same 16 readings fed, one step per frame, to an EKF started at 0 m (blue, follows the truth, black) and one started at 11 m (red): the red filter moves left while the robot moves right, wrong by up to 10.5 m with narrow error bars of two standard deviations, until the robot passes the beacon at 6 m (grey dotted line). The last frame shows the whole run, both filters ending near the truth

A **Gaussian** (G-827), one bell-shaped curve, is **unimodal** (G-2528), with a single mode (peak), so it cannot say "here or at the mirror place"; that belief has two modes, it is **multimodal** (G-1275), and a **particle filter** (G-2492), which stores the belief as a cloud of samples, can keep both until the readings decide (Labbe ch.12).

The other limits (Freiburg EKF slides; Labbe ch.11):

- **Strongly nonlinear models.** Where the Gaussian is wide compared with the bend (Section 3.3), the linearised covariance is wrong, usually too small, and the filter becomes

overconfident.

- **Jacobians must exist.** A model with steps, like the door map of the **particle filter** (G-2492) (1.0 m beside wall, 1.5 m at a door), has slope 0 almost everywhere and no slope at the edges, so the EKF cannot use it.
- **Jacobians must be derived.** Deriving them by hand is error-prone for big models; symbolic tools such as SymPy can do it (Labbe ch.11).

The unscented **Kalman filter** (G-2506) pushes a few chosen sample points through the nonlinear function instead of a tangent line, and the particle filter pushes many; both avoid Jacobians at a higher cost (Labbe ch.9; Freiburg EKF slides). Each of these limits can leave the EKF wrong while its variance stays small, so it needs a check.

5. Consistency: does the filter's variance match its real error?

> **Key point:** A filter is consistent when its real errors are as big as the variance it reports with its estimate: on average, error squared divided by variance comes to the number of variables. The **innovations** (G-2511) allow the check without knowing the truth.

5.1 Why we need a check

To detect a wrong result, the filter's own variance is of no use. The EKF's variance comes from tangent lines and from the noise variances we chose, the slip's Q and the sensor's R, 0.01 m^2 each, so it can be wrong without any error message: the mirror filter of Figure 5 reported about 10 cm of uncertainty while 10 m off. A controller or a planner that relies on a variance needs to know whether that variance matches the real errors. A filter whose errors match its own variance is **consistent** (G-2524) (Chen et al. 2018; Labbe ch.8). On a real robot the true errors are unknown, but the **innovations** (G-2511), the gaps between each reading and the reading the filter predicted, are not.

5.2 The innovation and its covariance

At every step, before the reading arrives, the filter predicts it. The filter also predicts how far the reading will be from that prediction. At step 1:

- $\bar{x}$ is the predicted position, 0.5 m;
- $h(\bar{x})$ is the reading the filter predicts, the beacon distance at $\bar{x}$, 5.852 m;
- z is the reading, 5.61 m;
- H is the measurement Jacobian, the slope of the beacon distance at $\bar{x}$, -0.940;
- $\bar{P}$ is the prediction's covariance matrix, here one number, the variance 1.01 m^2;
- H^T is the transpose of H, H flipped over its diagonal; for one number, H itself;
- R is the sensor's variance, 0.01 m^2;
- S is the innovation covariance, in m^2.

The **innovation** (G-2511) $z - h(\bar{x})$ has two independent sources of error, the prediction's and the sensor's. The prediction's error reaches the reading through the slope H, so its variance is multiplied by H^2 (variance times slope squared); then the two independent variances add:

$$S = H\bar{P}H^{\mathsf{T}} + R$$

This S is the **innovation covariance** (G-2523): the variance the filter expects of its innovation (Labbe ch.8; Chen et al. 2018). At step 1, the innovation was -0.242 m with $S = 0.902$ m^2, a predicted standard deviation of:

$$\sqrt{S} = \sqrt{0.902} = 0.95 \text{ m}$$

So an innovation of 0.24 m was well within what the filter expected. Unlike the true error, the innovation is known on a real robot at every step, so it is the routine test.

5.3 NIS: the innovation test, no truth needed

To turn each **innovation** (G-2511) into one test number, divide its square by its predicted variance. At step 1:

- y is the innovation, the reading minus the reading the filter predicted, -0.242 m;
- S is its predicted variance, the innovation covariance, 0.902 m^2;
- ε_z is the test number, with no unit.

$$\varepsilon_z = y^2/S$$

For several readings at once, y is a vector with one entry per reading, S is a matrix with one row and one column per reading, y^{T} is y written as a row, and dividing by S becomes multiplying by its inverse S^{-1}:

$$\varepsilon_z = y^{\mathsf{T}} S^{-1} y$$

The number ε_z is the **normalised innovation squared** (NIS) (G-2526). For step 1:

$$\varepsilon_z = 0.242^2/0.902 = 0.065$$

Why it should average 1. If the filter is **consistent** (G-2524), y follows a **Gaussian** (G-827), a bell-shaped curve, with mean 0 and variance S, so $y/\sqrt{S}$ is a standard normal variable (mean 0, variance 1) and its square averages 1. A squared standard normal variable follows the chi-square distribution (G-378) with 1 degree of freedom; with m readings at once, the NIS follows chi-square with m degrees of freedom and averages m (Chen et al. 2018). For one reading, 95 percent of a consistent filter's NIS values lie below 3.84 (notebook), and an innovation two standard deviations out gives an NIS of 4.

On one run. Figure 6 plots 60 innovations with the band $\pm 2\sqrt{S}$ the filter predicts. With the right sensor variance R, 0.01 m^2, 2 percent fall outside. A filter that assumes the sensor is twice as precise ($R = 0.0025$, a standard deviation of 0.05 m) predicts a narrower band,

about ±0.24 m instead of ±0.32 m, and 13 percent fall outside: its innovations are bigger than its S allows (Labbe ch.8).

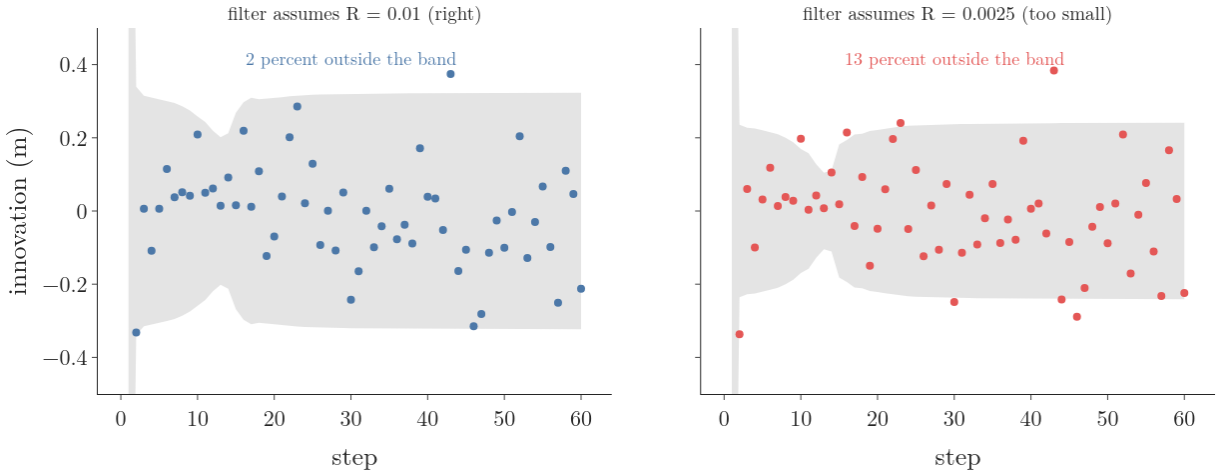


Figure 6: Innovations of one 60-step run with the band of two predicted standard deviations. With the right sensor variance R = 0.01 m^2, the band is about ±0.32 m and 2 percent fall outside; a filter that assumes R = 0.0025 m^2 predicts a band of about ±0.24 m, and 13 percent fall outside

5.4 NEES: the error test, when the truth is known

In a simulation, or with a much better reference sensor, the true state is known and we can test the error itself. Divide the squared error by the filter's variance. At step 1:

- x is the true position, 0.73 m;
- $\hat{x}$ is the filter's estimate, 0.755 m;
- p is the filter's variance, 0.0112 m^2;
- ε_x is the test number, with no unit.

$$\varepsilon_x = (x - \hat{x})^2 / p$$

For a state of n numbers, the error $\tilde{\mathbf{x}}$, the true state minus the estimate, is a vector of n entries, P is the filter's covariance matrix, n rows by n columns, and dividing by P becomes multiplying by its inverse P^{-1}:

$$\varepsilon_x = \tilde{\mathbf{x}}^{\mathsf{T}} P^{-1} \tilde{\mathbf{x}}$$

The number ε_x is the **normalised estimation error squared** (NEES) (G-2525) (Labbe ch.8; Chen et al. 2018). By the same argument it follows chi-square with n degrees of freedom and averages n, here 1. For step 1:

$$\varepsilon_x = (0.73 - 0.755)^2 / 0.0112$$

$$= 0.056$$

One value says little; the test averages many. We simulated 500 runs of 16 steps, ran three EKFs on each, set with different slip variances Q and sensor variances R, and averaged NEES and NIS at each step (Figure 7). If the filter is **consistent** (G-2524), an average of 500 chi-square values with 1 degree of freedom lies between 0.88 and 1.13 in 95 percent of cases (notebook):

Filter	Average NEES	Average NIS	Verdict
right noises, $Q = 0.01$, $R = 0.01$	1.00	1.01	consistent
R too small, 0.0025	3.25	2.41	overconfident
Q too small, 0.0001	15.95, rising to 32.9	2.12	overconfident, getting worse

In Figure 7 the grey band marks 0.88 to 1.13: only the filter with the right noises stays inside it, the NEES of the filter with too small an R stays near 3, and the NEES of the filter with too small a Q rises past 30.

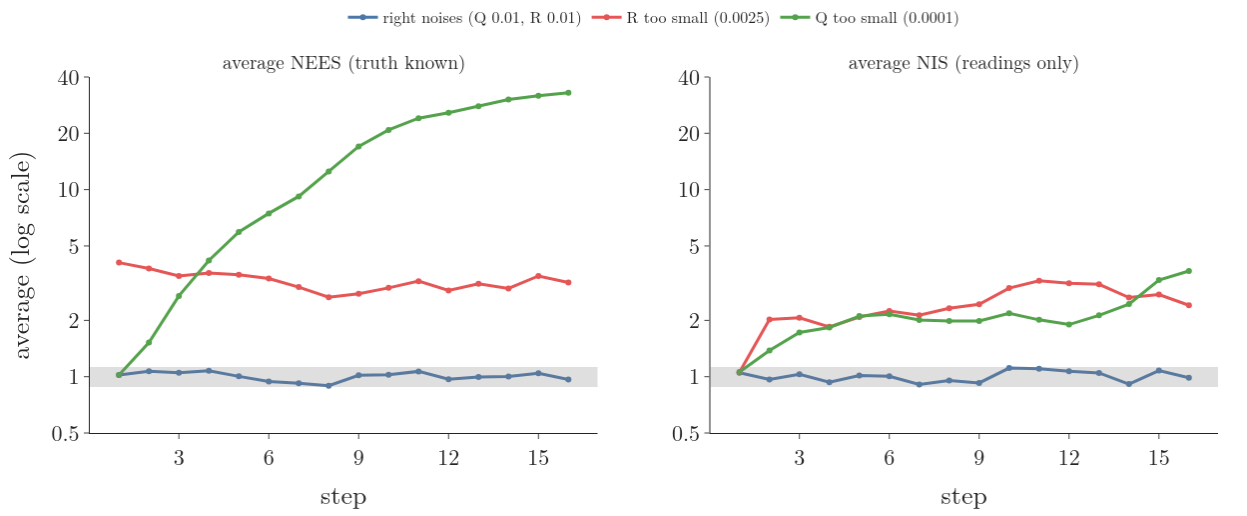


Figure 7: Average NEES (left) and NIS (right) over 500 runs, per step, log scale. The filter with the right noises stays in the grey band around 1; the filter with too small an R stays near 3; the one with too small a Q has an NEES that rises past 30

5.5 Reading the results

- **Both near 1:** the filter's variances match its real errors; it can be trusted.
- **Both well above 1:** the filter is overconfident. Increase the noise it assumes; which one is told by tuning experiments such as those of choosing Q and R.
- **NEES rising over time:** errors accumulate faster than the filter's variance allows, the mark of too small a slip variance Q: the filter relies on its motion model and gives the readings almost no weight.
- **Values well below 1:** the filter is underconfident; its noises are too large, and it wastes information.

The **innovations** (G-2511) should also look like random noise: centred on zero and with no slow trend. A trend means the model omits something, such as a bias or too little process noise (Labbe ch.8). NEES needs the truth, so it is a design-time test in simulation; NIS needs only the readings, so it can also run on the robot.

6. Summary

Idea	What it does	Why
Linearisation	replace a nonlinear function by its tangent line at the estimate	a linear function keeps a Gaussian a Gaussian
First-order propagation	sd out $\approx$ slope $\times$ sd in	the tangent line's slope stretches the Gaussian
EKF	Kalman filter with the motion Jacobian G and the measurement Jacobian H recomputed every step	nonlinear models with unimodal beliefs
Innovation covariance S	the variance the filter expects of its innovation	a yardstick for each reading
NIS, NEES	squared innovation or error over its predicted variance, average 1 per variable	tests whether the filter is consistent

- The beacon's distance is a **nonlinear function** (G-2530) of the position (2 m at its lowest, 2.83 m at both 4 m and 8 m), yet the EKF still locates the robot from it, ending the 16 steps at 7.749 m against a true 7.86 m, because it straightens the curve at every prediction.
- A nonlinear function bends a Gaussian: near the beacon the readings concentrate just above 2 m with **skewness** (G-1817) 1.89, so a mean and variance alone no longer describe them.
- Linearising at the mean is almost exact where the curve is nearly straight across the **Gaussian** (G-827)'s width (sd 0.46 both ways at 1 m) and too narrow where it bends (tangent-line sd 0.36 against the true 0.57 for a wide Gaussian at 5.5 m), so EKF accuracy depends on keeping the uncertainty small compared with the bends.
- The EKF uses the nonlinear functions for the means and their slopes for the covariances, so with linear models it is exactly the **Kalman filter** (G-2506).
- The slope sets how much information a reading carries, because the position is known only to within the sensor's standard deviation divided by the slope: at 0.94 that is 0.106 m; under the beacon the slope is near 0 and the filter relies on its commands.
- One Gaussian holds one hypothesis, so an EKF started on the wrong side follows the mirror path, 10.5 m off with narrow error bars; **multimodal** (G-1275) beliefs need a particle filter.
- A **consistent filter** (G-2524)'s NEES and NIS average 1 per variable (1.00 and 1.01 in our runs); values near 3 or a rising NEES flag noises set too small, and NIS works without the truth, so it can run on the robot.

The robot can now locate itself with a sensor that measures a distance, not a position, and check with NIS whether it is consistent, as long as the belief is **unimodal** (G-2528). Still missing is a close look at the sensors that give these readings and their noise, the subject of the next chapter, robot sensors.

7. Sources

Built from

- MATLAB, "Understanding Kalman Filters, Part 5: Nonlinear State Estimators", YouTube, https://www.youtube.com/watch?v=Vefia3JMeHE
- James Han, "The Extended Kalman Filter (EKF): Why Taylor Expansions are Awesome", YouTube, https://www.youtube.com/watch?v=9X3jGGnbcvU
- Lars Hammarstrand (Chalmers), "4.3.2 Kalman filter tuning and consistency: Innovation", YouTube, https://www.youtube.com/watch?v=QQLO7N3PQkM
- Lars Hammarstrand (Chalmers), "4.3.1 Kalman filter tuning and consistency", YouTube, https://www.youtube.com/watch?v=E5JEWjNITxQ
- Labbe, R. *Kalman and Bayesian Filters in Python*, ch.8 "Designing Kalman Filters" (NEES, likelihood, residuals), ch.9 "Nonlinear Filtering", ch.11 "Extended Kalman Filters", ch.12 "Particle Filters", https://github.com/rlabbe/Kalman-and-Bayesian-Filters-in-Python (Labbe ch.8, ch.9, ch.11, ch.12)
- Becker, A. *Kalman Filter from the Ground Up*, ch.13 "Extended Kalman Filter (EKF)": uncertainty projection, EKF equations, limitations; tutorial page https://www.kalmanfilter.net/ekf.html (Becker ch.13)
- Burgard, W. et al. (Uni Freiburg), *Introduction to Mobile Robotics*, "Bayes Filter – Extended Kalman Filter" slides, http://ais.informatik.uni-freiburg.de/teaching/ss23/robotics/slides/11-ekf.pdf (Freiburg EKF slides)
- Chen, Z., Heckman, C., Julier, S. and Ahmed, N. (2018). "Weak in the NEES?: Auto-tuning Kalman Filters with Bayesian Optimization". arXiv:1807.08855, §II (definitions of NEES and NIS and their chi-square tests), https://arxiv.org/abs/1807.08855 (Chen et al. 2018)

Other references

- NPTEL-NOC IITM, "Introduction to Robotics, #34 Extended Kalman Filter", YouTube, https://www.youtube.com/watch?v=c7V5_uQ2ues
- Thrun, S., Burgard, W. and Fox, D. (2005). *Probabilistic Robotics*. MIT Press. §3.3 "The Extended Kalman Filter"; cited for what the Freiburg slides and Labbe confirm (Thrun)
- Bar-Shalom, Y., Li, X. R. and Kirubarajan, T. (2001). *Estimation with Applications to Tracking and Navigation*. Wiley. The origin of the NEES and NIS tests, cited by Chen et al. 2018

8. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Nonlinear function (G-2530)	A function whose graph is not a straight line, so it is not a fixed multiple of the input plus a constant, such as the beacon distance $h(x) = \sqrt{(6-x)^2 + 4}$; it bends a Gaussian into another shape, which is why the Kalman filter cannot use it as it is and the extended Kalman filter linearises it.
First-order uncertainty propagation (G-2519)	Carrying a mean and a standard deviation through a function by its value and slope at the mean: mean out $\approx h(\mu)$, standard deviation out $\approx$ slope $\times$ standard deviation in (with a Jacobian, JPJ^{T}); accurate when the function is nearly straight across the standard deviation.
Motion Jacobian G (G-2522)	The table of slopes of the motion model at the last estimate, one row per new state variable and one column per old one; the EKF uses it to carry the covariance forward as $GPG^{\mathsf{T}} + Q$.
Extended Kalman filter (EKF) (G-2520)	The Kalman filter for nonlinear motion and sensor models: it moves the mean through the nonlinear functions and uses their Jacobians at the latest estimate in place of F and H, so it can track unimodal beliefs through nonlinear models.
Measurement Jacobian H (G-2521)	The slopes of the sensor model at the predicted state, one row per reading and one column per state variable; the EKF uses it to compute the innovation covariance and the gain, and its size says how much information a reading carries about the state.
Consistent filter (G-2524)	A filter whose real errors are as large as its own reported covariance says, on average; only then can its uncertainty be trusted by a controller or planner.
Innovation covariance S (G-2523)	The variance the filter expects of its next innovation, $S = H\bar{P}H^{\mathsf{T}} + R$: prediction uncertainty seen through the sensor plus sensor noise; it scales the gain and is the yardstick for the NIS test.

Term	Meaning
NIS (normalised innovation squared) (G-2526)	The squared innovation divided by its predicted covariance, $y^\mathsf{T} S^{-1} y$; for a consistent filter it follows chi-square with m degrees of freedom (one per reading) and averages m, so it tests consistency from readings alone, even on the robot.
NEES (normalised estimation error squared) (G-2525)	The squared estimation error divided by the filter's variance, $\tilde{x}^\mathsf{T} P^{-1} \tilde{x}$; for a consistent filter it follows chi-square with n degrees of freedom and averages n, so its average over simulated runs tests consistency when the truth is known.
Innovation (G-2511)	The gap between a reading and the reading the prediction expected, $z - H\bar{x}$; the update moves the estimate by the gain times this gap, and its size is used to check the filter.
Kalman filter (G-2506)	The Bayes filter for a linear Gaussian system: it keeps the belief as a mean and a (co)variance and updates them with a predict step and an update step, blending a model's prediction with a sensor reading so the result is as certain as possible.
Gaussian distribution (G-827)	Another name for the normal distribution: the symmetric, bell-shaped continuous distribution set by its mean and standard deviation, used to model many measurements.
Linear function (G-2527)	A function $y = ax + b$: a fixed multiple of the input plus a constant, whose graph is a straight line; passing a normal variable through it gives another normal variable, which is why the Kalman filter is exact for linear models.
Kalman gain K (G-2508)	The weight the update gives to the reading, $K = \bar{p}/(\bar{p} + R)$ in one dimension: the weight that makes the updated variance smallest, near 1 when the prediction is vague and near 0 when the reading is.
Measurement noise R (G-2510)	The random error of a sensor reading, given as its variance or covariance R; it sets how much weight the filter gives each reading and is taken from the datasheet or measured in a calibration.

Term	Meaning
Skewness (G-1817)	A number for how lopsided a distribution is: 0 symmetric, positive right tail, negative left tail.
Linearisation (G-1099)	Replacing a function near a point by its tangent line (its first-order Taylor polynomial), so that a curve can be handled as a straight line close to that point, where straight-line methods apply.
Derivative (G-595)	The slope of a function at a point: how fast the output changes as the input changes there. It is zero at the lowest point of a smooth curve, which is how training finds a minimum.
Chain rule (G-371)	To differentiate a function of a function, multiply the outer derivative by the inner derivative.
Pose (G-2280)	Where a robot is and which way it faces: on a floor, its position (x, y) and heading θ, such as (2 m, 1 m, 30°); needed because a robot at one spot drives off differently depending on its heading.
Jacobian (G-980)	The table (matrix) of all first partial derivatives of a function with several inputs and outputs, one row per output and one column per input, $J_{ij} = \partial f_i / \partial x_j$; it shows how every output changes with every input near a point.
Unimodal (G-2528)	Having a single mode (one peak); a unimodal belief can be held by one Gaussian, so the Kalman filter needs only its mean and variance.
Multimodal (G-1275)	Having more than one mode, that is more than one peak (two peaks: bimodal), such as a robot's belief peaking at three doors; it shows the data point to several separate values, so one mean or one best guess would hide them.
Particle filter (G-2492)	A Bayes filter that stores the belief as a set of weighted samples (particles) and runs each round as move, weigh, resample; it can hold beliefs of any shape, such as a belief with three modes, one at each of three doors.

Term	Meaning
Chi-square distribution (G-378)	The shape the chi-square statistic χ^2 follows when the null hypothesis H_0 is true: never negative, skewed to the right, with one parameter, the degrees of freedom.